

## Assignment nr. 5 – Antenna arrays and BFN

**Assigned:** Tuesday May 12<sup>th</sup>, 2026  
**Due:** Wednesday May 27<sup>th</sup>, 2026 at class starting

### Notes

To take the exam with the grading of submitted assignments + oral discussion it is necessary:

- to solve **individually** the exercises;
- to submit the solution **within the deadline**, uploading the PDF in the section “Elaborati” of the web portal

### Problem nr. 1

Using the script prepared in class, compute and plot the array factor  $AF(\alpha)$  (in dB) as a function of the observable angle  $\alpha$  for a generic linear array in the following cases.

- Equispaced and uniform broadside array with  $N = 7$  elements and  $d/\lambda = 1 - 1/N$ .
- Equispaced and uniform broadside array with  $N = 6$  elements and  $d/\lambda = 8.67$ .
- Equispaced broadside array with  $N = 7$  elements,  $d/\lambda = 0.8$  and  $\mathbf{A} = [a_1, a_2, \dots, a_7] = [0.44 \ 0.734 \ 0.936 \ 1.0 \ 0.936 \ 0.734 \ 0.44]$ .
- Equispaced broadside array with  $N = 7$  elements,  $d/\lambda = 0.8474$  and  $\mathbf{A} = [a_1, a_2, \dots, a_7] = [0.544 \ 0.694 \ 0.916 \ 1.0 \ 0.916 \ 0.694 \ 0.544]$ .

For a better comparison among the different cases, plot the array factors in the same plot.

### Problem nr. 2

Design a broadside array directed along  $x$ , whose elements are elementary electric dipoles directed along  $y$ , satisfying the following conditions:

- total width of the main beam  $\text{FNBW} = \Delta\beta \leq 30^\circ$ ;
- no grating lobes.

Determine the phase shift  $\phi$ , the minimum number of elements  $N$  and the distance between two elements  $d/\lambda$ . Draw the (qualitative) radiation pattern of the whole antenna (i.e. taking into account also that of the dipole) in the three principal planes.

### Problem nr. 3

Design an array directed along  $z$ , whose elements are elementary electric dipoles directed along  $x$ , satisfying the following conditions:

- direction of maximum radiation along  $+z$ ;
- total width of the main beam  $\text{FNBW} = \Delta\alpha \leq 100^\circ$ ;
- no grating lobes.

Determine the phase shift  $\phi$ , the minimum number of elements  $N$  and the distance between two elements  $d/\lambda$ . Draw the (qualitative) radiation pattern of the whole antenna (i.e. also considering the dipole) in the three principal planes.

### **Problem nr. 3**

Design a corporate BFN to feed a broadside array of 4 patches as that designed in problem 1 of homework 4 (use the value of  $Z_{in}(f_0) = R_0$  you found).

The BFN access line must have an equivalent characteristic impedance  $Z_0 = 50 \Omega$ .

The substrate is FR4 material ( $\epsilon_r = 4.45$ ,  $h = 1.55$  mm). The strip widths must be no less than 0.15 mm (0.175 mm minimum spacing between distinct tracks) to avoid fabrication problems and excessive losses. The strip width should not exceed 5 mm to avoid spurious radiation.

Draw the BFN, indicating the width and the length of each line section.

Data:  $f_0 = 5.8$  GHz